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CAREER-FOCUSED PROFESSIONAL TRAINING

DevOps with AI

3-Month Course | 2 Hours per Day | Monday to Friday

Career roles: DevOps Engineer • Cloud Engineer

Course Overview

Build practical knowledge of Linux, Cloud Computing, CI/CD, Infrastructure as Code, Containerization, Orchestration, Automation and Monitoring through real-world projects. AI tools are used in every module to help generate, explain, debug and improve DevOps work.

Duration	Schedule	Class time
3 months	Monday to Friday	2 hours per day

What You Will Learn

- Linux administration, command-line operations, users, permissions, services, logs and troubleshooting.
- Bash scripting, Git and GitHub workflows, and CI/CD automation with Jenkins and GitHub Actions basics.
- AWS cloud services, networking, access management, monitoring and security best practices.
- Containerization with Docker and application orchestration with Kubernetes.
- Infrastructure as Code using Terraform and server configuration automation using Ansible.
- Monitoring and alerting with Prometheus, Node Exporter, Alertmanager and Grafana.
- Practical use of AI to generate and debug scripts, YAML, Dockerfiles, Jenkinsfiles, Terraform code and PromQL.

LEARNING APPROACH | Learn the concept • Build by hand • Use AI • Verify the output • Explain your solution

Module-wise Curriculum

The course combines foundational concepts with hands-on implementation and AI-assisted troubleshooting.

MODULE	TOPICS / LEARNING FOCUS
1. DevOps Fundamentals + AI for DevOps	DevOps lifecycle; Waterfall vs Agile; Agile and Scrum basics; monitoring concepts; prompting for DevOps; verifying AI output.
2. Linux	Installation; commands; users and groups; permissions; processes; packages; systemctl; disk partitioning; SSH; networking commands; cron and at jobs; logs and log rotation; troubleshooting with AI; website deployment project on Ubuntu.
3. Bash Scripting	Variables; conditions; loops; functions; arrays; case statements; file handling; AI script generation and debugging; automation scripts.
4. Git & GitHub	Core commands; branching; merge; rebase; stash; tags; conflicts; pull requests; GitHub Actions basics; AI for commit messages and workflows.
5. AWS	IAM; EC2; S3; VPC (subnets, route tables, IGW, NAT); security groups; NACL; ALB; Auto Scaling; CloudWatch; CloudTrail; RDS; Route 53; AWS CLI; security best practices.
6. Jenkins & CI/CD	Installation; freestyle and pipeline jobs; Jenkinsfile; Git/Maven/Docker integration; triggers; plugins; backup; security; end-to-end pipeline; AI for Jenkinsfiles and build logs.

Module-wise Curriculum — Continued

MODULE	TOPICS / LEARNING FOCUS
7. Docker	Images and containers; Dockerfile; volumes; networking; Compose; multi-stage builds; Docker Hub; security basics; best practices; AI for Dockerfiles.
8. Kubernetes	Architecture; Minikube; kubectl; Pods; Deployments; Services; ConfigMaps; Secrets; PV/PVC; Ingress; rolling updates and rollbacks; StatefulSets; DaemonSets; Helm; RBAC; troubleshooting with AI.
9. Terraform	Installation; providers; variables; outputs; state and remote backend; modules; workspaces; EC2, VPC and S3 builds.
10. Ansible	Inventory; ad-hoc commands; playbooks; variables; handlers; roles; Galaxy; server configuration; multi-server automation.
11. Monitoring	Prometheus; Node Exporter; Alertmanager; Grafana dashboards; basic alerts; AI for PromQL.
12. Real-Time Projects	Four hands-on projects designed to apply the course tools and workflows in realistic scenarios.

Real-Time Projects

Project 1: Website deployment on Ubuntu

Deploy and verify a website on an Ubuntu environment.

Project 2: CI/CD pipeline using Jenkins, Git and Docker

Connect source control, build automation and containerization in an end-to-end pipeline.

Project 3: Dockerized application deployed on Kubernetes

Package an application in Docker and deploy it to Kubernetes.

Project 4: AWS infrastructure built with Terraform and configured with Ansible

Provision cloud infrastructure as code and automate server configuration.

Week-wise Schedule

WEEK	TOPICS
1	DevOps fundamentals + AI introduction; Linux
2	Linux
3	Bash scripting; Git
4	Git & GitHub; AWS
5	AWS
6	AWS; Jenkins
7	Jenkins; Docker
8	Docker; Kubernetes
9	Kubernetes
10	Terraform
11	Ansible; Monitoring
12	Monitoring; real-time projects

How We Use AI in This Course

• **Generate:** scripts, YAML, Dockerfiles, Jenkinsfiles and Terraform code.

• **Debug:** paste an error or log into an approved AI tool, ask for an explanation and possible fix, then verify the fix.

• **Explain:** use AI to understand commands and policies, then explain them back in your own words.

• **Build fundamentals first:** do the first lab of each module by hand, then use AI.

• **Own your work:** be able to explain every AI-generated output you use.

• **Protect sensitive data:** never paste secrets, access keys or company data into AI tools.

Practice Outside Class

Plan for 30 to 60 minutes of daily practice on your own AWS free-tier account, VM or Minikube. Regular hands-on practice is the key to finishing this course job-ready.

Career pathways
DevOps Engineer • Cloud Engineer